Lancaster University Leipzig

Inheritance vs Composition in OOP

LZMSCI521M | Week 1

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Why Object-Oriented Programming?

1. We model the world as objects: **state** and **behaviour**.
2. Class = Mould, a **blueprint** for object; it defines **attributes** and **methods**.
3. Why it matters: **encapsulation, reuse, polymorphism**.

Today you will learn:

1. Objects vs Classes, state vs behaviours, and examples in `Java`, `Python`, and `UML`.
2. Relationship between classes: Inheritance vs Composition

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OOP Concepts

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Notion Of Classes: Basic Syntax

Mould for building Objects:

1. **Attributes**: state representation, characteristics
2. **Methods**: behaviors, state controller.

```java
public class Rectangle{

  Pair<Integer, Integer> p1 = new Pair<Integer, Integer>(1, 2);
  Pair<Integer, Integer> p2 = new Pair<Integer, Integer>(3, 1);

  // Methods
  public int area() {
    int width = Math.abs(p1.getValue0() - p2.getValue0());
    int height = Math.abs(p1.getValue1() - p2.getValue1());
    return width * height;
  }
}
```

Objects

- Objects are instances of a class
- Objects can also represent different states of a class
- Objects are coherent entities that store data and the code (or instructions) working on that data.

Constructor

Example

```
Rectangle x = new Rectangle()
x.area()
```

```
Pair<Integer, Integer> p1 = Pair.with(5,2);
Pair<Integer, Integer> p2 = Pair.with(2, 3);
Rectangle y = new Rectangle(p1, p2);
```

Instantiation (Constructors)

```java
public class Rectangle{
  Pair<Integer, Integer> p1 = Pair.with(1, 2);
  Pair<Integer, Integer> p2 = Pair.with(3, 1);

  //Constructor
  public Rectangle(Pair<Integer, Integer>p1, Pair<Integer,
   Integer> p2) {
    this.p1= p1;
    this.p2 = p2;
  }
  // Methods
  public int area() {
    int width = Math.abs(p1.getValue0() - p2.getValue0());
    int height = Math.abs(p1.getValue1() - p2.getValue1());
    return width * height;
  }}
```

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Inheritance vs Composition in OOP

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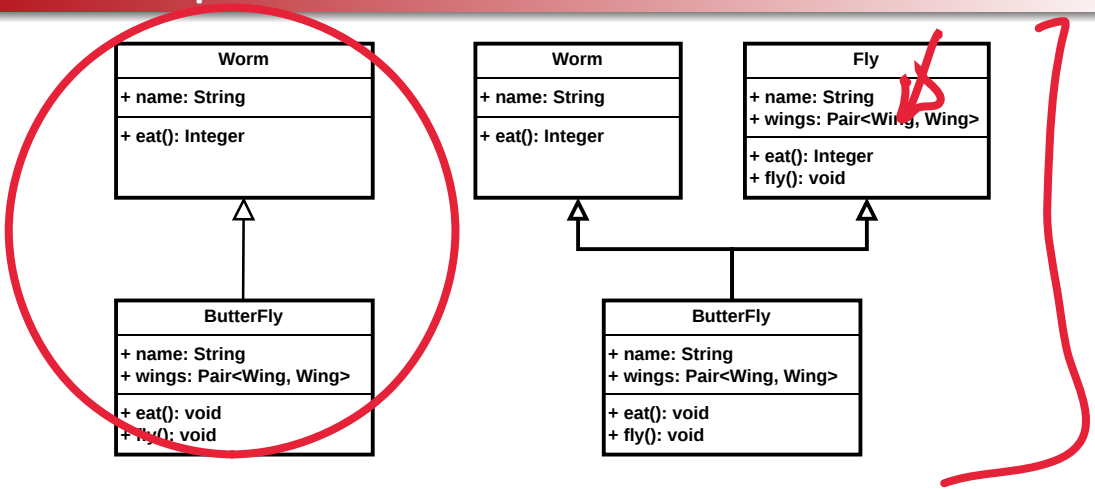
OOP: Inheritance (is-a)

- A subclass inherit from a base class to **reuse** and **specialise** its parents' attributes and methods.
- Classes inherited from another are derived classes, **subclasses**, or **subtypes**.
- Classes from which other classes are derived are called base classes or **superclasses**.
- Use when the subclass truly **is-a** base class and respects the **LSP** (Liskov Subtitution Principle).

```java
public class Square extends Rectangle {}
```

```java
Square square = new Square();
System.out.println(square); //Rectangle([1, 2], [3, 1])
```

OOP: Multiple inheritance



- Conflict of method names
- Repetition or duplication of information

OOP: Multiple inheritance and Java

A class can be derived from more than one superclass in Python.

```
class Worm:
  def __init__ (self, name):
    self.name = name
  def eat(self):
    print(self.name +" swallows")
class Fly:
  def __init__ (self, name):
    self.name = name
  def eat(self):
    print(self.name +" is nibbling..")
class ButterFly(Worm, Fly):
  pass
```

bf = BF()

bf.eat()

??

In Java: How can we implement it? (Home work (1)) ??

OOP: Composition (has-a, delegation)

Composition is an OOP concept that models the relationship **has-a**.

- **Composite class**: **has a** components and **delegates** work to them.
- **Component class**: part of the composite class

Example: We already used it in the previous Example, but how and where? (Home work (2))

Remarks

- More **flexible** compared to inheritance
- Changes to the component class have minimal or no effects on the composite class
- Better **encapsulation**, but a bit more wiring.

Example of Composition in Java

```java
public class Fly {
  public String name;
  public Pair<Wing, Wing> wings;

  public Fly(String name, Wing left, Wing right) {
    this.name = name;
    this.wings = Pair.with(left, right);
  }

  public void eat() {
    System.out.println(this.name +" is nibbling...");
  }
  public void fly() {
    System.out.println("Hi! I believe I can fly. ");
  }}
```

Example of Composition in Java

```java
public class Wing {
  public String name = null;
  public Integer length = 4;
  public String wingColor = "white";

  public Wing() {}

  public Wing(String name, Integer length, String color) {
    this.name = name ;
    this.length = length;
    this.wingColor = color;
  }
}
```

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Takeaway

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Composition vs Inheritance (when to choose which)

Inheritance (is-a)

- *Pros*: Simple reuse, polymorphism via base type.
- *Cons*: Tight coupling to base; fragile base class; LSP risks.
- *Use when*: True subtype, stable base API, shared behaviour.

Composition (has-a)

- *Pros*: Replaceable parts, multiple change axes, better encapsulation.
- *Cons*: More wiring; no automatic access to base internals.
- *Use when*: Need variability, combine features, avoid deep trees.

Remarks: Prefer composition for flexibility; use inheritance for true subtyping with stable behaviour.

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Home Work 1.0

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Questions

1. Consider the Multiple inheritance UML class diagram on **Slide 18**:
   a. Use the nation of Interface to implement that diagram
   b. How can we solve the conflict of method names in `Python`?
2. Given the `Java` Class Rectangle in **Slide 11**:
   a. Find where the composition is being used
   b. Can we replace that by an Inheritance relationship?
   c. How can this example illustrate the polymorphism?
3. When would **inheritence** still be the cleaner choice?

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What Next?

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What next?

Encapsulation, Polymorphism, and other Class relationships, e.g., association, aggregation, etc...

Beyond OOP

1. Domain Driven Design (DDD)
2. Agent Oriented Programming (AOP)
3. Design Paterns

Some Important Materials

- Bloch, Joshua. Effective Java. 3rd ed. Addison-Wesley, 2018.
- Gamma, Erich; Helm, Richard; Johnson, Ralph; Vlissides, John. Design Patterns: Elements of Reusable Object-Oriented Software. Addison-Wesley, 1994.
- Code and Course Material